Yuxuan Liu

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Education

University of Virginia 09/2025 – present

Ph.D. Candidate in Computer Engineering | GPA: **4.00**/4.00

Core Courses: Sensors and Ubiquitous Computing (A), Neural Networks (A), Computer Architecture and Design (A+)

Tsinghua University 09/2021 – 07/2024

Master of Science in Data Science and Information Technology | GPA: **3.98**/4.00; Ranking: **8**/134

Core Courses: Advanced Signal Processing: Methods and Practice (A+), Nanogenerators and Self-powered Systems (A-), Operations Research (A-), Fundamentals of Information Theory (A), Course Project of Embedded Systems (A)

Beijing Jiaotong University 09/2017 - 07/2021

Bachelor of Engineering in Communications Engineering | GPA: **3.86**/4.00; Ranking: **7**/254

Core Courses: Signals and Systems (97/100), Digital Signal Processing (94/100), Digital Electronics Technology (96/100), Fundamentals of Wireless Communications (90/100), Principles of Communication Systems (92/100), Electromagnetic Field and Magnetic Waves (94/100)

Publications

[C12] Haoran Zhang, **Yuxuan Liu**, Patricia D. Flores, Michael C. Clark, Elizabeth C. Courtney, Robert Theile, Emily E. Grammer, Kyle C. Quinn, Macy Stahl, Arthur L. Weltman, Jason D. Allen, Amanda Watson, *"PMST: A Physiologically Mediated Spectral Transformer for Wearable Blood Lactate Estimation"*, **ACM SenSys'27** (Under Review)

[C11] **Yuxuan Liu**, Amanda Watson, Macy Stahl, Emily Grammer, Arthur L. Weltman, Jason D. Allen, Haoran Zhang, Patricia D. Flores, *"VO_2ptic: Continuous VO_2 Monitoring via Wearable Multispectral Optical Sensing"*, **ACM SenSys'27** (Accepted)

[C10] **Yuxuan Liu**, Lex Fedor, Chaitali Harge, Beth Hundt, Bill Shuttleworth, Andrew Carlson, Amanda Watson, *"Machine Learning-Based Real-Time Detection of Spreading Depolarizations Using Continuous Multi-Physiology Monitoring"*, **COSBID iCSD'26**

[J4] **Yuxuan Liu***, Nan Zhou*, Haoyang Wang, Fanhang Man, Jingao Xu, Fan Dang, Chaopeng Hong, Yunhao Liu, Xiao-Ping Zhang, Xinlei Chen, *"CatUA: Catalyzing Urban Air Quality Insights through Mobile Crowd-Sensing"*, **IEEE TMC**, 2025 (CCF-A, TH-CPL-A, SCI-Q1, DOI)

[J3] Rongye Shi*, Ji Luo*, Nan Zhou, **Yuxuan Liu**, Chaopeng Hong, Xiao-ping Zhang, Xinlei Chen, "Phy-APMR: A. physics-informed air pollution map reconstruction approach with mobile crowd-sensing for fine-grained measure-ment", Elsevier Building and Environment, 2025 (SCI-Q1, DOI)

[C9] **Yuxuan Liu***, Haoyang Wang*, Fanhang Man, Jingao Xu, Fan Dang, Yunhao Liu, Xiao-Ping Zhang, Xinlei Chen, *"MobiAir: Unleashing Sensor Mobility for City-scale and Fine-grained Air-Quality Monitoring with AirBERT"*, **ACM MobiSys'24** (Acceptance Rate≈16.3%, DOI)

[C8] Weichen Zhang, **Yuxuan Liu**, Xuzhe Wang, Xuecheng Chen, Chen Gao, Xinlei Chen, *"Demo Abstract: Embodied Aerial Agent for City-level Visual Language Navigation Using Large Language Model"*, **ACM/IEEE IPSN'24** (DOI)

[C7] Haoyang Wang, Xinyu Luo, Ciyu Ruan, Xuecheng Chen, Wenhua Ding, **Yuxuan Liu**, Xinlei Chen, *"Poster: Fusing Event and Depth Sensing for Dynamic Objects Localization and Tracking"*, **ACM HotMobile'24** (DOI)

[C6] Yuhan Cheng, Xuecheng Chen, Yixuan Yang, Haoyang Wang, **Yuxuan Liu**, Xinlei Chen, *"Poster: Olfactory Sensing in Turbulent Airflow via Collaborative Robots"*, **ACM HotMobile'24** (DOI)

[C5] Chenyu Zhao, Ciyu Ruan, Shengbo Wang, Jirong Zha, Haoyang Wang, Jiaqi Li, **Yuxuan Liu**, Xuzhe Wang, Xinlei Chen, *"Demo Abstract: Bio-inspired Tactile Sensing for MAV Landing with Extreme Low-cost Sensors"*, **ACM/IEEE IPSN'24** (DOI)

[J2] Xuecheng Chen, Haoyang Wang, Yuhan Cheng, Haohao Fu, **Yuxuan Liu**, Fan Dang, Yunhao Liu, Jinqiang Cui, Xinlei Chen, *"DDL: Empowering Delivery Drones with Large-scale Urban Sensing Capability"*, **IEEE JSTSP**, 2024 (DOI)

[C4] **Yuxuan Liu***, Haoyang Wang*, Chenyu Zhao, Jiayou He, Wenbo Ding, Xinlei Chen, *"CaliFormer: Leveraging Unlabeled Measurements to Calibrate Sensors with Self-supervised Learning"*, **ACM UbiComp/ISWC'23** (CPD Workshop, DOI)

[C3] Haoyang Wang, Fanhang Man, Zihan Wang, **Yuxuan Liu**, Xinlei Chen, Wenbo Ding, *"Poster Abstract: TENG-enabled Self-powered Human-machine Interfaces for the Metaverse"*, **ACM/IEEE IPSN'23** (DOI)

[C2] **Yuxuan Liu***, Yifei Sun*, Ziteng Wang, Xiaoli Qu, Dezhi Zheng, Xinlei Chen, *"C-RIDGE: Indoor CO2 Data Collection System for Large Venues Based on Prior Knowledge"*, **ACM SenSys'22** (DATA Workshop, DOI)

[C1] **Yuxuan Liu**, Xinyu Liu, Fanhang Man, Chenye Wu, Xinlei Chen, *"FAD: Fine-grained Air Pollution Data Enables Smart Living and Efficient Management"*, **ACM SenSys'22** (DOI)

[J1] **Yuxuan Liu**, *"Design of an Internet of Things (IoT)-based shipping monitoring platform and database system"*, **The Journal of New Industrialization**, 2020,10(03):126-128+133 (ISSN 2095-6649)

Conference Presentation

- Oral Presentation: ML-based Real-time SD Detection, COSBID iCSD'26, Charlottesville, U.S., September 2026
- Oral Presentation: MobiAir, ACM MobiSys'24, Tokyo, Japan, June 2024
- Oral Presentation: CaliFormer, ACM UbiComp/ISWC'23, Cancún, Mexico, October 2023
- Oral Presentation: C-RIDGE, ACM SenSys'22, Virtual, November 2022
- Demo Presentation: FAD, ACM SenSys'22, Virtual, November 2022

Research Experience & Projects

[R1] **Wearable Broadband Optical Sensing for Noninvasive Biomarker Monitoring** 09/2025 - present

Role: Project Leader & Principle Researcher | Adviser: Prof. Amanda Watson

➢ ***VO_2ptic: Continuous VO_2 Monitoring via Wearable Multispectral Optical Sensing***

- Propsoed VO2ptic, a wrist-worn multispectral optical sensing system for continuous and low-cost VO_2 monitoring.
- Developed a physics-informed ratio-based feature extraction framework to mitigate tissue scattering, anatomical variability, and common-mode motion interference that obscure metabolic spectral signatures.
- Designed a two-phase VO_2 prediction framework combining a SVR to capture nonlinear mapping between spectral features and VO_2 with a customized Resformer to model temporal dependencies and correct prediction residuals.
- Validated VO2ptic system on **11** subjects across **14** exercise-lab sessions with LOSO-CV, achieving R^2 of **0.62**, MAE of **5.22mL/kg/min**, and **1.51ms** latency, which outperformed SOTA baselines by **0.45** on R^2 and **27.0%** on MAE.

➢ ***PMST: A Physiologically Mediated Spectral Transformer for Wearable Blood Lactate Estimation***

- Proposed a wearable multispectral optical sensing system for noninvasive blood lactate estimation, addressing the weak optical signatures and inter-subject variability within optical signals.
- Designed a physiology-mediated spectral learning framework to extract lactate-related information from multi-wavelength optical measurements while accounting for physiological variations during exercise.
- Developed a hierarchical lactate estimation pipeline to model both within-session lactate dynamics and between-subject variability, enabling reconstruction of absolute blood lactate levels from wearable optical measurements.
- Validated the system through in-vitro optical characterization and an in-vivo cycling study involving **11** participants and **20** exercise sessions, achieving R^2 of **0.623**, RMSE of **1.096mmol/L**, and MAE of **0.867mmol/L**.

[R2] **Real-Time Detection of Spreading Depolarization (SD)** 06/2026 - present

Role: Project Leader & Principle Researcher | Adviser: Prof. Amanda Watson & Prof. Andrew Carlson

➢ ***Machine Learning-based Real-Time Detection of SDs Using Continuous Multi-Physiology Monitoring***

- Developed a machine learning framework for real-time SD detection using routinely acquired HR, SpO_2, SBP, DBP, and MAP signals, aiming to enable continuous SD monitoring independent of invasive ECoG-based methods.
- Designed changing-rate feature engineering methods to normalize physiological changes over unequal time intervals due to non-uniform sampling during SD/non-SD class balancing, enabling capturing physiological dynamics.
- Benchmarked **13** ML models spanning MLP, KNN, SVM, tree-based, and linear approaches on data from **73** subjects, with GradientBoosting achieving **0.904** accuracy, **0.941** precision, **0.923** recall, and **0.932** F1-score for patient-level diagnosis, where changing-rate features further improved accuracy and F1-score by **34.7%** and **18.6%**, respectively.
- Demonstrated the robust SD detection performance at a **10-minute** temporal resolution from 1-60 minutes, with KNN achieving accuracy over **0.864**, F1-score over **0.887**, and recall over **0.932**.

[R3] **RISC-V CPU Design** 01/2026-04/2026

Role: Project Leader | Adviser: Prof. Todd DeLong

- Designed a complete RV32I **single-cycle CPU** from scratch, including the full datapath and control logic to support **6** types of instructions (R-, I-, S-, B-, U-, and J-type).
- Extended the single-cycle architecture into a **5-stage pipelined CPU**, partitioning instruction execution into instruction fetch, decode, execute, memory access, and write-back stages for overlapped instruction execution.
- Implemented pipeline hazard-resolution mechanisms, including register forwarding, pipeline stalls, and pipeline flushing to address data dependencies, load-use hazards, and control hazards, respectively.
- Verified both CPU architectures using ModelSim and RARS through customized assembly-level testing, with both implementations successfully passing all test cases and reducing execution time by **4.10×** for the single-cycle CPU

and **2.96×** for the pipelined CPU compared with typical multi-cycle implementation.

[R4] **City-scale and Fine-grained Air Quality Sensing** 09/2021 – 06/2024

Role: Project Leader | Adviser: Prof. Xinlei Chen

➢ ***CaliFormer: Leveraging Unlabeled Measurements to Calibrate Sensors with Self-supervised Learning***

- Proposed CaliFormer, a sensor calibration model based on representation learning to extract temporal- and spatial-invarient knowledge shared among tasks to address the calibration issue in environmental data.
- Pioneered the integration of self-supervised learning into sensor calibration tasks to address the challenge of limited labels in environmental data.
- Introduced a Transformer-based model architecture and designed a Masked Language Modeling (MLM) pre-training task to train the calibration model effectively.
- The MAE of data calibrated by CaliFormer can be reduced to **16.54** $\mu g/m^3$, achieving a **30%** improvement compared to the SOTA methods. Furthermore, the model exhibited strong robustness under different labeling rates.

➢ ***MobiAir: Unleashing Sensor Mobility for City-scale and Fine-grained Air-Quality Monitoring with AirBERT***

- Proposed MobiAir, the first accurate city-scale and fine-grained air quality monitoring system based on MCS.
- Implemented a modular sensing front-end with separate chambers and active airflow design, which aims to collect highly mobile air quality data under high sampling rates.
- Developed AirBERT to separate the spatially mixed gas measurements by extracting non-stationary and non-Markovian features and modeling the mutual influences among sensor readings.
- Designed a knowledge-informed training strategy to enhance performance of MobiAir and reduce the demand for city-scale labeled data.
- City-wide deployement and experimental results demonstrate MobiAir reduces sensing errors by **96.7%** with only **44.9ms** latency, outperforming the SOTA baseline by **39.5%**.

➢ ***FAD: Fine-grained Air Pollution Data Enables Smart Living and Efficient Management***

- Proposed FAD, a multi-source environmental data-driven system, using fine-grained air pollution data to realize smart city.
- Designed a low-cost and high-precision portable sensing device to collect high-spatial-resolution air pollution data. It can be deployed on highly mobile vehicles (e.g., buses, drones) to perform Mobile Crowdsensing (MCS) tasks.
- Designed four services: Advice for healthy life, Route planning, Toxic gas leakages finding & warning, and High pollution-emission sources locating to improve residential living quality and official city management efficiency.
- Collaborate with Environmental Thinking Technology Co.,Ltd to deploy over **300** devices in **17** cities and regions across China, with a total annual device operating time exceeding **1.5 million** hours and a data volume exceeding **2 billion** records. Experiments show that the data spatial resolution can be up to **45m**, representing an improvement of over **40 times** compared to traditional static sensing methods.

[R5] **Smart Sensing and Energy Conversion Based on Self-powered TENG** 09/2021 - 02/2023

Role: Project Leader | Adviser: Prof. Wenbo Ding

➢ ***TENG-enabled Self-powered Human-machine Interfaces for the Metaverse***

- Proposed a human-machine interface (HMI) system based on the soft self-powered TENG (Triboelectric Nanogenerator) to address the limited degrees of freedom issue encountered in traditional HMIs.
- Assist in implementing an ergonomically designed TENG sensor prototype deployed on a human joint to collect electrical signals generated during body movements (e.g., the arm), and utilizing simple Artificial Neural Networks (ANN) to reconstruct and visualize the angles of motion.
- The recognition resolution of bending angle of human arms can be up to **15°**, and our system can map real movements in the digital world.

➢ ***Distributed Low-frequency Mechanical Energy Conversion based TENG Prototype***

- Designed standalone layered TENG sensors with low-cost acrylic sheets, simulated wave motion, collected sensor data with electrostatic meter, and analyzed signal waveforms using LabVIEW.
- Investigated the impact of materials, contact area, and movement speed on signal intensity by altering substrate materials, roller dimensions, and sensor tilt angles, and enhanced signal through stacking multiple layers of sensors.
- Varified effectiveness of converting low-frequency mechanical energy into electrical energy by illuminating LEDs.

[R6] **Low-Carbon Winter Olympics - Intelligent Data Collection and Processing** 09/2021 - 10/2022

Role: Group Leader | Advised by Prof. Xinlei Chen & Prof. Dezhi Zheng

➢ ***C-RIDGE: Indoor CO_2 Data Collection System for Large Venues Based on Prior Knowledge***

- Proposed C-RIDGE, a low-power sensing system for high-resolution CO_2 data collection in large venues.
- Proposed a power optimization and anomaly detection module based on prior physical knowledge in sensor design, and develop a sensor deployment strategy driven by data and physical models as the deployment method.
- Collaborate with BOCOG to deploy **45** C-RIDGE systems in the National Speed Skating Oval during the 2022 Beijing Winter Olympics, and carried out **15-day** data collection of CO_2 concentrations with data released on Zenodo.
- Experiments demonstrated that C-RIDGE achieved a **36.1%** reduction in power consumption. The anomaly detection module achieved a **100%** detection rate for anomalies and effectively identified abnormal trends.

Working Experience

➢ **Internship**

- LLM Development Intern @ Meituan Co.,Ltd 05/2025-07/2025
- System Research & Development Intern @ Environmental Thinking Technology Co.,Ltd 01/2024-06/2024
- Research Assistant @ Shenzhen Institute of Artificial and Robotics for Society (AIRS) 09/2021-10/2022

➢ **Service**

• USENIX OSDI (2026)	Artifact Evaluation	05/2026
• IEEE EMBC (2026)	Invited Reviewer	02/2026
• ACM SIGCOMM (2025)	Artifact Evaluation	09/2025
• ACM TOSN (2025, 2026)	Invited Reviewer	09/2025, 08/2026
• IEEE JSTSP (2023)	Invited Reviewer	11/2023
• ACM UbiComp/ISWC(CPD Workshop, 2023)	Session Chair	10/2023

Selected Honors

2025 | **President Fellowship**, University of Virginia
Distinguished Fellowship, University of Virginia
2024 | **Outstanding Thesis** (M.Sc.) of Tsinghua University & **Outstanding Graduate** of Beijing City
ACM MobiSys Student Travel Grants
Dean's Scholarship, Tsinghua University
First place of the Best Poster, International Workshop on Learning and Information Theory
Third prize of TBSI Retreat Student Poster Contest, Tsinghua University
2023 | **The Award of BEST PRESENTATION in CPD Session**, ACM UbiComp/ISWC
Second-class Overall Excellence Scholarship, Tsinghua University
2017-2021 | **First-class Academic Excellence Scholarship**, Beijing Jiaotong University (top 3%)
First-class Arts Excellence Scholarship, Beijing Jiaotong University (top 5%)
Second-class Sports Excellence Scholarship, Beijing Jiaotong University (top 5%)
Merit Students, Beijing Jiaotong University (top 3%)
Honorable Mention of Mathematical Contest in Modeling/Interdisciplinary Contest in Modeling (MCM/ICM)
Third prize at the College Student Computer Design Contest

Skills

Programming Language: Python, Matlab, Embedded C, Assembly Language, Verilog, Latex, MySQL, Java, HTML, JSP
Software & Platform: VS Code, Arduino, Pycharm, Eclipse, Multisim, Tina-TI, Protues, Keil5, Wireshark, NS2, Quartus